

Name: _____

Date: _____

PYTHAGORAS, (FINDING THE HYPOTENUSE)

LO: I can use Pythagoras' theorem to find the hypotenuse of a right-angled triangle.

Finding the hypotenuse

The **hypotenuse** is the longest side of a right-angled triangle, opposite the right angle. Use $c^2 = a^2 + b^2$ then square root.

Strategy: Square both shorter sides, add them together, then take the square root to find c.

Quick examples

●	a = 3, b = 4: c = $(9 + 16) = 25$ = 5	3-4-5 triple
●	a = 5, b = 12: c = $(25 + 144) = 169 = 13$	5-12-13 triple
●	a = 6, b = 8: c = $(36 + 64) = 100 = 10$	6-8-10 triple
●	c = a + b = 3 + 4 = 7	must square, add, then root

Key idea

Square both shorter sides, add them, then square root to find the hypotenuse.

$$c = (\text{a}^2 + \text{b}^2)$$

Hypotenuse = (sum of squares of shorter sides)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - Pythagorean triple

Find the hypotenuse when a = 6 and b = 8.

$$c^2 = 6^2 + 8^2 = 36 + 64 = 100$$

$$c = 100$$

$$c = 10$$

$$c = 10$$

6, 8, 10 is a Pythagorean triple (multiple of 3, 4, 5).

Example 2 - non-integer answer

Find the hypotenuse when a = 5 and b = 7.

$$c^2 = 5^2 + 7^2 = 25 + 49 = 74$$

$$c = 74$$

$$c = 8.60 \text{ (2 d.p.)}$$

$$c = 8.60$$

Leave as 74 for exact form, or round for decimal.

Example 3 - in context

A field has sides 40 m and 30 m at right angles. Find the diagonal.

$$c^2 = 40^2 + 30^2 = 1600 + 900 = 2500$$

$$c = 2500$$

$$c = 50 \text{ m}$$

$$50 \text{ m}$$

The diagonal of a right-angled arrangement is the hypotenuse.

Example 4 - with decimals

Find the hypotenuse when a = 2.5 and b = 6.

$$c^2 = 2.5^2 + 6^2 = 6.25 + 36 = 42.25$$

$$c = 42.25$$

$$c = 6.5$$

$$c = 6.5$$

Decimal inputs follow the same process: square, add, root.

YOUR TURN – PRACTICE (deliberate practice)

1. Integer answers (triples)

Find the hypotenuse.

- a) $a = 3, b = 4$
- b) $a = 5, b = 12$
- c) $a = 8, b = 15$
- d) $a = 9, b = 40$

2. Decimal answers

Find c to 1 d.p.

- a) $a = 4, b = 6$
- b) $a = 3, b = 7$
- c) $a = 5, b = 9$
- d) $a = 2, b = 8$

3. Word problems

Find the hypotenuse in each situation.

- a) Walk 6 km East then 8 km North. Direct distance?
- b) Gate 3 m wide, 4 m tall. Diagonal brace length?
- c) TV screen 48 cm wide, 36 cm tall. Diagonal?
- d) Triangle: base 7 cm, height 24 cm. Hypotenuse?

4. Larger numbers

Find the hypotenuse. Give exact or rounded answer.

- a) $a = 20, b = 21$
- b) $a = 11, b = 60$
- c) $a = 14, b = 48$
- d) $a = 10, b = 10$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $c^2 = a^2 + b^2$ then $c = (a^2 + b^2)$

☐

Correct answer: _____

Reason: _____

b) The hypotenuse equals $a + b$

☐

Correct answer: _____

Reason: _____

c) The hypotenuse is always the longest side ☐

Correct answer: _____

Reason: _____

d) $c = (a + b)$ ☐

Correct answer: _____

Reason: _____

e) 3, 4, 5 and 5, 12, 13 are Pythagorean triples ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A right-angled triangle has legs (x) and ($x + 1$). The hypotenuse is ($x + 2$). Use Pythagoras to find x . Show this gives the 3-4-5 triangle.

Expression: _____

Simplified: _____

b) A ship sails 24 km due North, then 7 km due East, then 10 km due North. Find the direct distance from start to finish.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

